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Wave Equation FEM Solver

Discretization Choices, Dissipation/Dispersion, and Numerical Results

Qiansong Li Chengxi Rao Chenyu Ye Zhenyu Zhao

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Outline

1. Problem and Model
2. Numerical Methods
3. Implementation and Experiments
4. Numerical Results

Project Goal and PDE

Goal: build and analyze a 2D FEM solver for the scalar wave equation, and discuss time/space discretization choices in terms of **accuracy, dissipation, and dispersion**.

Initial-Boundary Value Problem

$$\begin{cases} u_{tt} - \Delta u = f, & (x,t) \in \Omega \times (0,T), \\ u = g, & (x,t) \in \partial\Omega \times (0,T), \\ u(x,0) = u_0(x), \quad u_t(x,0) = v_0(x), & x \in \Omega. \end{cases}$$

- Benchmark in code: eigenmode manufactured solution on unit square.
- Additional case: driven boundary to study physical energy injection.
- Implementation: deal.II + simplex meshes.

Weak Form and Space Semi-Discretization

For homogeneous Dirichlet BC, with $V = H_0^1(\Omega)$:

$$(u_{tt}, v) + a(u, v) = (f, v), \quad a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx.$$

Choose FEM space $V_h = \text{span}\{\phi_i\}_{i=1}^N$, $u_h = \sum_j U_j(t)\phi_j$.

Matrix ODE System

$$M\ddot{\mathbf{U}}(t) + K\mathbf{U}(t) = \mathbf{F}(t),$$

$$M_{ij} = \int_\Omega \phi_i \phi_j \, dx, \qquad K_{ij} = \int_\Omega \nabla\phi_i \cdot \nabla\phi_j \, dx.$$

- This is the common starting point for all tested time integrators.
- Mass treatment (consistent/lumped) changes both cost and dynamics.

Non-homogeneous Dirichlet BC Handling

For $u = g(x,t)$ on $\partial\Omega$, introduce a lifting:

$$u(x,t) = w(x,t) + \ell(x,t), \qquad \ell|_{\partial\Omega} = g, \quad w|_{\partial\Omega} = 0.$$

Then for all $v \in H_0^1(\Omega)$:

$$(w_{tt}, v) + a(w, v) = (f, v) - (\ell_{tt}, v) - a(\ell, v).$$

After FEM discretization, split interior/boundary DoFs:

$$\begin{bmatrix} M_{II} & M_{IB} \\ M_{BI} & M_{BB} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{U}}_I \\ \ddot{\mathbf{g}}_B \end{bmatrix} + \begin{bmatrix} K_{II} & K_{IB} \\ K_{BI} & K_{BB} \end{bmatrix} \begin{bmatrix} \mathbf{U}_I \\ \mathbf{g}_B \end{bmatrix} = \begin{bmatrix} \mathbf{F}_I \\ \mathbf{F}_B \end{bmatrix}.$$

Hence the interior equation to advance is

$$M_{II}\ddot{\mathbf{U}}_I + K_{II}\mathbf{U}_I = \mathbf{F}_I - M_{IB}\ddot{\mathbf{g}}_B - K_{IB}\mathbf{g}_B.$$

The boundary trace $\mathbf{U}_B(t) = \mathbf{g}_B(t)$ is prescribed data.

Central Difference Method

Discretize acceleration:

$$\ddot{\mathbf{U}}^n \approx \frac{\mathbf{U}^{n+1} - 2\mathbf{U}^n + \mathbf{U}^{n-1}}{\Delta t^2},$$

giving

$$\mathbf{U}^{n+1} = 2\mathbf{U}^n - \mathbf{U}^{n-1} - \Delta t^2 M^{-1} K \mathbf{U}^n.$$

- Explicit update when M^{-1} is cheap (best with lumped mass).
- CFL-type stability condition:

$$\Delta t \leq \frac{2}{\sqrt{\lambda_{\max}(M^{-1}K)}}.$$

- In stable range: low dissipation, but phase error accumulates (numerical dispersion).

Newmark-β Method

Kinematic update:

$$\mathbf{U}^{n+1} = \mathbf{U}^n + \Delta t\, \mathbf{V}^n + \Delta t^2 \left(\tfrac{1}{2} - \beta\right) \mathbf{A}^n + \beta \Delta t^2 \mathbf{A}^{n+1},$$

$$\mathbf{V}^{n+1} = \mathbf{V}^n + \Delta t (1-\gamma) \mathbf{A}^n + \gamma \Delta t\, \mathbf{A}^{n+1}.$$

Using $M\mathbf{A}^{n+1} + K\mathbf{U}^{n+1} = \mathbf{F}^{n+1}$:

$$\left(M + \beta \Delta t^2 K\right) \mathbf{U}^{n+1} = M\mathbf{U}^\star + \beta \Delta t^2 \mathbf{F}^{n+1}.$$

- One sparse linear solve per step (implicit).
- Used in project: $(\beta, \gamma) = (1/4, 1/2)$.
- Central difference can be viewed as a special explicit member of the Newmark family.

Mass Matrix Choices and Four Combinations

Consistent mass

$$M_{ij} = \int_\Omega \phi_i \phi_j \, dx$$

- Galerkin-consistent
- Sparse but non-diagonal
- More expensive in explicit-like updates

Lumped mass

$$M_{ii}^{\text{lump}} = \sum_j M_{ij}$$

- Diagonal approximation
- Very cheap inverse
- Practical choice for explicit wave runs

Compared combinations: central difference + lumped/consistent, Newmark + lumped/consistent.

Implementation Architecture

- `src/Wave2D.cpp`: mesh reading, assembly, BC treatment, time stepping, diagnostics, VTU/CSV output.
- `src/main.cpp`: CLI interface for mesh, Δt, T, time scheme, mass type, FE degree, etc.
- `scripts/run_experiments.sh`: batch experiments.
- Plot scripts: `plot_convergence.py`, `plot_energy_comparison.py`, `plot_error_comparison.py`, `dispersion_analysis.py`.

Typical workflow

build $\rightarrow$ run batch cases $\rightarrow$ generate CSV/VTU $\rightarrow$ produce report figures.

Experiment Matrix and Metrics

- Exact benchmark solution for error evaluation:

$$u(x, y, t) = \sin(\pi x)\sin(\pi y)\cos(\omega t), \quad \omega = \sqrt{2}\pi.$$

- **Spatial refinement:** vary h with small fixed Δt.
- **Temporal refinement:** vary Δt on fine fixed mesh.
- **Scheme/mass comparison:** same $h, \Delta t$, different methods.

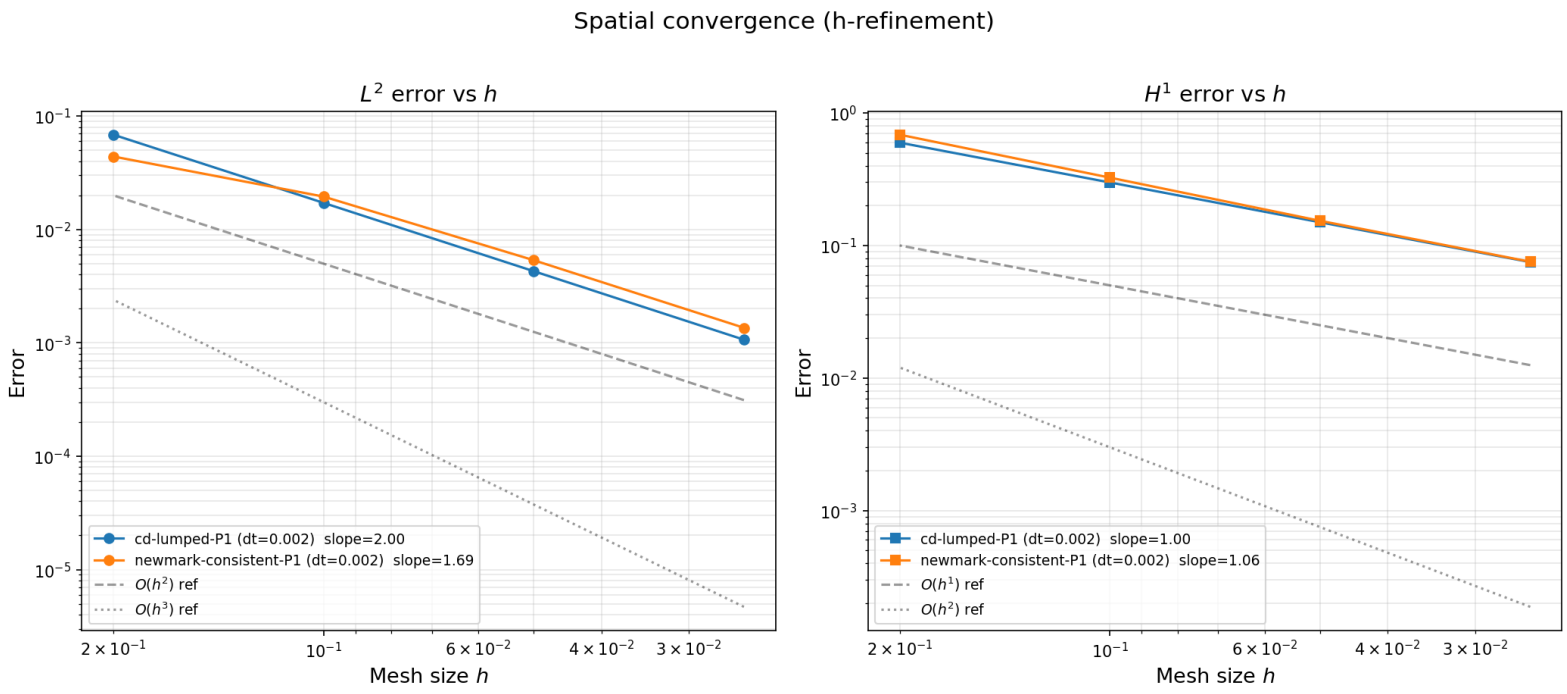
Monitored outputs

- Error: L^2, H^1 vs time and at final time.
- Discrete energy:

$$E(t) = \frac{1}{2}\mathbf{v}^T M\mathbf{v} + \frac{1}{2}\mathbf{u}^T K\mathbf{u}.$$

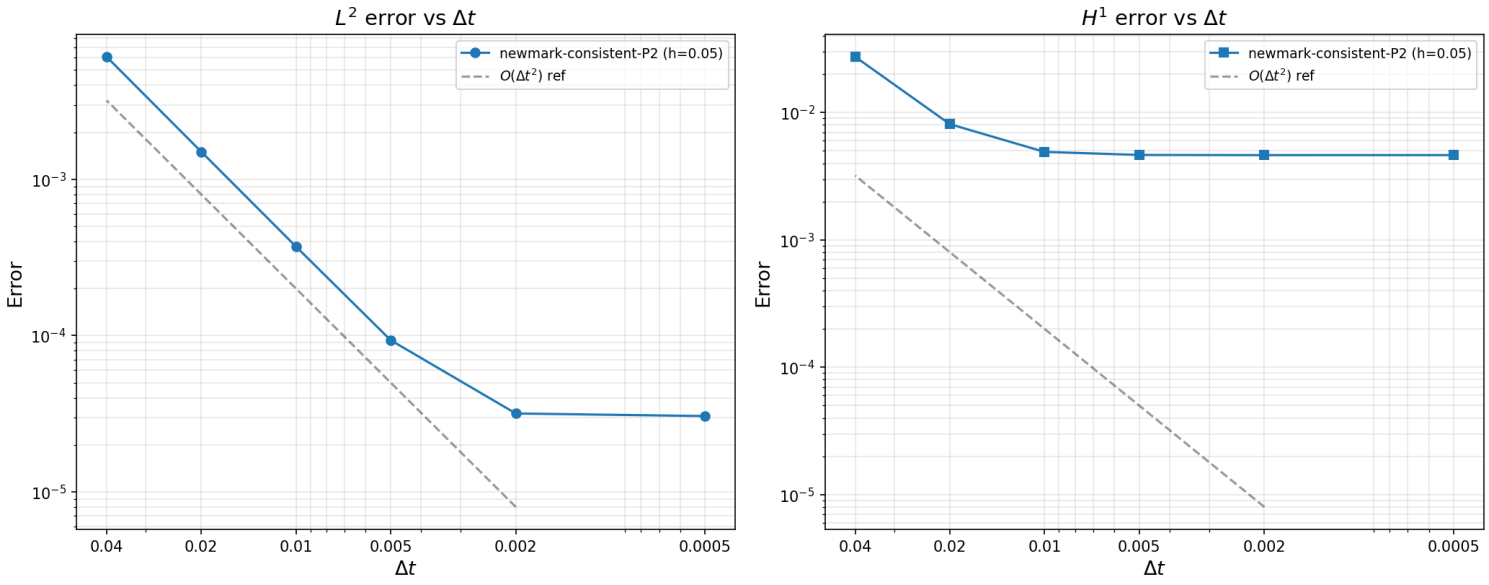
- Runtime statistics (mean/std/median).

Spatial Convergence



- Error decreases monotonically under mesh refinement.
- For smooth eigenmode benchmark, trend is close to expected second-order behavior in L^2 with P1 elements.

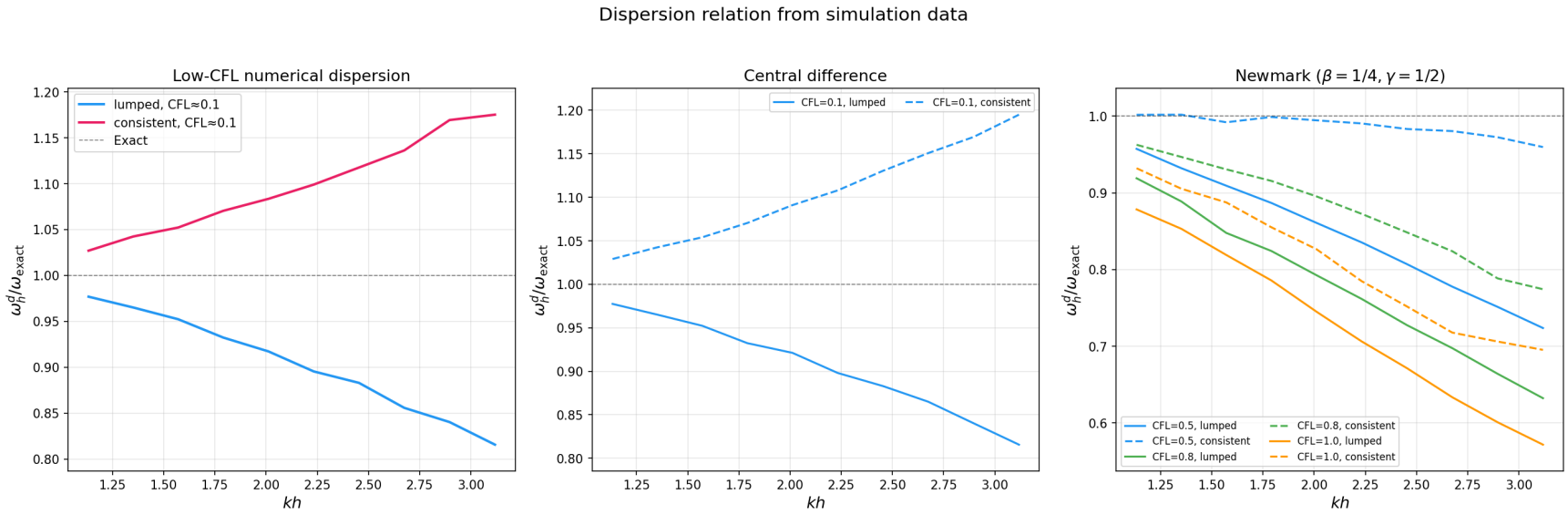
Temporal Convergence



- Reducing Δt lowers time error until a spatial-error floor appears.
- Setup used here: Newmark ($h = 0.05,\ P2,$ consistent).
- This is consistent with

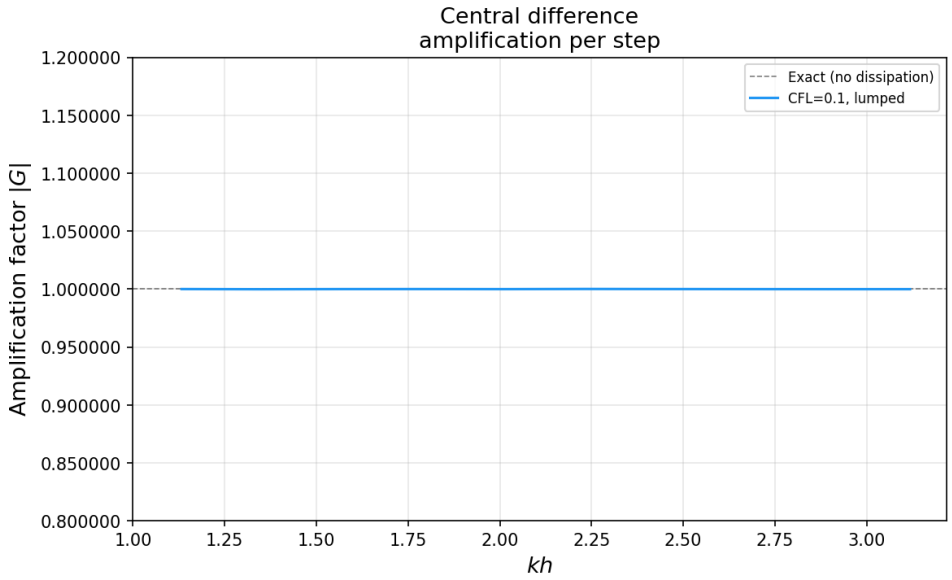
$$\|u - u_h\| \lesssim C_1 h^{p+1} + C_2 \Delta t^2.$$

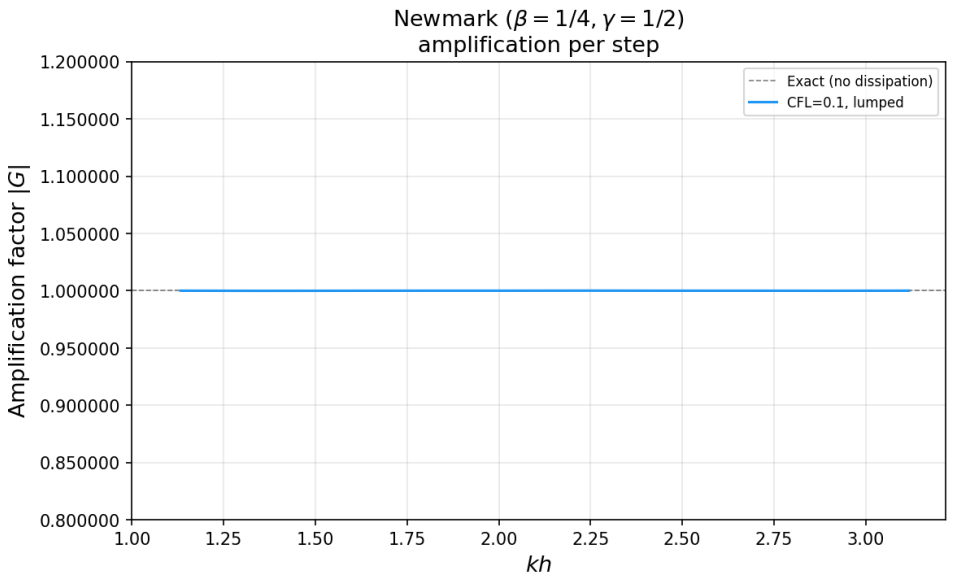
Dispersion Diagnostic



- Axis meaning: horizontal axis is kh (non-dimensional wavenumber), vertical axis is $\omega_h^d/\omega_{\text{exact}}$ (phase-speed ratio).
- The ratio $\omega_h^d/\omega_{\text{exact}}$ departs from 1 at large normalized wavenumber.
- This means short waves travel with the wrong phase speed, producing accumulated phase error over long integrations.

Dissipation Diagnostic (Amplification Factor)





- Axis meaning: horizontal axis is kh (non-dimensional wavenumber), vertical axis is $|G|$ (amplitude amplification per time step).
- For central difference and Newmark $(\beta, \gamma) = (1/4, 1/2)$, curves stay close to $|G| = 1$ in the stable range.
- Hence the dominant numerical effect is dispersion rather than strong artificial damping.

Runtime Comparison

Method + Mass	mean [s]	std [s]	median [s]	rel.
central diff. + lumped	0.0154	0.0184	0.0083	1.00x
central diff. + consistent	0.0523	0.0051	0.0548	6.59x
Newmark + lumped	0.0466	0.0095	0.0431	5.19x
Newmark + consistent	0.0530	0.0075	0.0543	6.53x

- Fastest: central difference + lumped mass.
- Consistent mass increases cost due to repeated sparse solves.
- Newmark variants are costlier because of implicit linear solve each step.